Troy Mayer

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Robotics software and controls: robot kinematics, state estimation, and real-time control in C++, built with tests and simulation. Mechanical engineering plus applied statistics; currently on Atlas at Boston Dynamics.

EDUCATION

Purdue University, School of Mechanical Engineering and School of Science — West Lafayette, IN
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics — May 2028
GPA 3.32/4 | Clubs: Club Swimming, House Band

EXPERIENCE

Boston Dynamics — Waltham, MA
Sensors and Perception Co-Op, Atlas — Jul 2026 – Present

- Kinematics: computing limb position and velocity at impact with floating-base forward kinematics (quaternions, Rodrigues' rotations, rigid-body velocity composition) from joint encoders, and mapping impacts onto the robot's mesh.
- Dynamics: estimating impact energy from Khatib's operational-space effective mass and an OLS velocity fit, and building a physics-based MuJoCo impact simulation.
- Tooling: built an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering ~28k impact events from ~900 robot test logs (ETL into SQLite, 3D impact heatmap, per-event waveforms), used by engineers during testing.
- Statistics: modeling hardware thermal behavior in Python, fitting heating curves by multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Caterpillar (via Purdue Data Mine) — West Lafayette, IN
Machine Learning Researcher — Jan 2026 – May 2026

- Built Python pipelines consolidating five years of legacy data and LSTM/XGBoost models forecasting next-day paint consumption with Monte Carlo confidence intervals, achieving a 4.7% reduction in material usage and contributing to a 3.3% cut in cycle time.

Chortos Lab, Soft Adaptive Systems and Structures — West Lafayette, IN
Undergraduate Researcher — Aug 2025 – May 2026

- Wrote a Python computer-vision tracker resolving displacement to 0.01 mm and built LabVIEW/MATLAB data acquisition for 50,000+ point datasets, guiding gel formulations that increased force output by over 50%.

ROBOTICS PROJECTS

Legged Robot State Estimation and Control | *Personal Project* — Aug 2026 – Oct 2026

- Built a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a simulated Unitree Go1: 0.013 m/s velocity RMSE over a 60 s walk vs. 1.38 m/s IMU-only, at ~20 µs per update.
- Proved it before trusting it: Jacobians checked against finite differences in GoogleTest, C++ matched to a Python reference within 1e-13, and an observability analysis showing why position and yaw drift while roll, pitch, and velocity stay bounded.
- Closed the loop: walking and 500 Hz QP force-balance controllers run on the estimate with 0 falls in 5 × 60 s walks, after tracing a 4° lean to weak tilt observability and fixing it with a 4 s startup turn.

Active Orthosis, Purdue ASME Biomechanics | *Electronics Team Lead* — Apr 2024 – May 2026

- Designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3, tuned from a first-order motor model with feed-forward and anti-windup, protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian EMG estimator, validated in closed-loop simulation on real recorded EMG with 26 GoogleTest tests in CI.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad (INA333, right-leg drive, ISO7763 isolation), raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot | *Independent Research, Purdue University & UC3M (Madrid)* — Oct 2026 – Present

- Designing push-off control optimized to hit a target launch speed, knee impedance control for landing, and reaction-wheel control to reorient mid-air for a three-legged hopping robot, simulated in MuJoCo.

SKILLS

- **Programming:** C++17, Python, MATLAB, SQL | Languages: English, Turkish (fluent), Spanish (fluent)
- **Robotics and controls:** Kalman filtering (error-state EKF), sensor fusion, rigid-body kinematics, PI and feed-forward control, QP force control, MuJoCo, Eigen, FreeRTOS
- **Software, data, hardware:** Linux, Git, GitHub Actions CI, CMake, GoogleTest, Flask, Plotly, SQLite, Pandas, SciPy, PyTorch | KiCad, LabVIEW, CATIA, NX, Creo